

AI, Machine Learning and Data Science



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Review of Previous Lecture

- Motivation behind Data Science
- Objectives and Outcome of the Course
- Operations of Data Science

Contents

- Data Science vs Machine Learning
- AI vs Data Science
- AI vs Machine Learning vs Data Science

Data Science and Machine Learning

Data Science

- Data science
 - Interdisciplinary field
 - Includes
 - Scientific methods,
 - Processes and
 - Algorithms on structured and unstructured data to extract insights.
- Data Science combines:
 - Mathematics,
 - Statistics,
 - Computer science, and
 - specific expertise to uncover patterns, trends, and relationships within data.

Data Science Stages

- Data collection.
- Data Storage
- Data preprocessing.
- Visualizing Data.
- Modeling and evaluation.

Machine Learning

- Machine learning, a subfield of artificial intelligence.
- Machine learning algorithms detect patterns and correlations within dataset.
- Machine learning systems can make complex predictions and analyses.
- The fundamental principle of Machine Learning lies in its autonomous learning capability.

Machine learning and Data Science (Similarities and Differences)

- Similarities

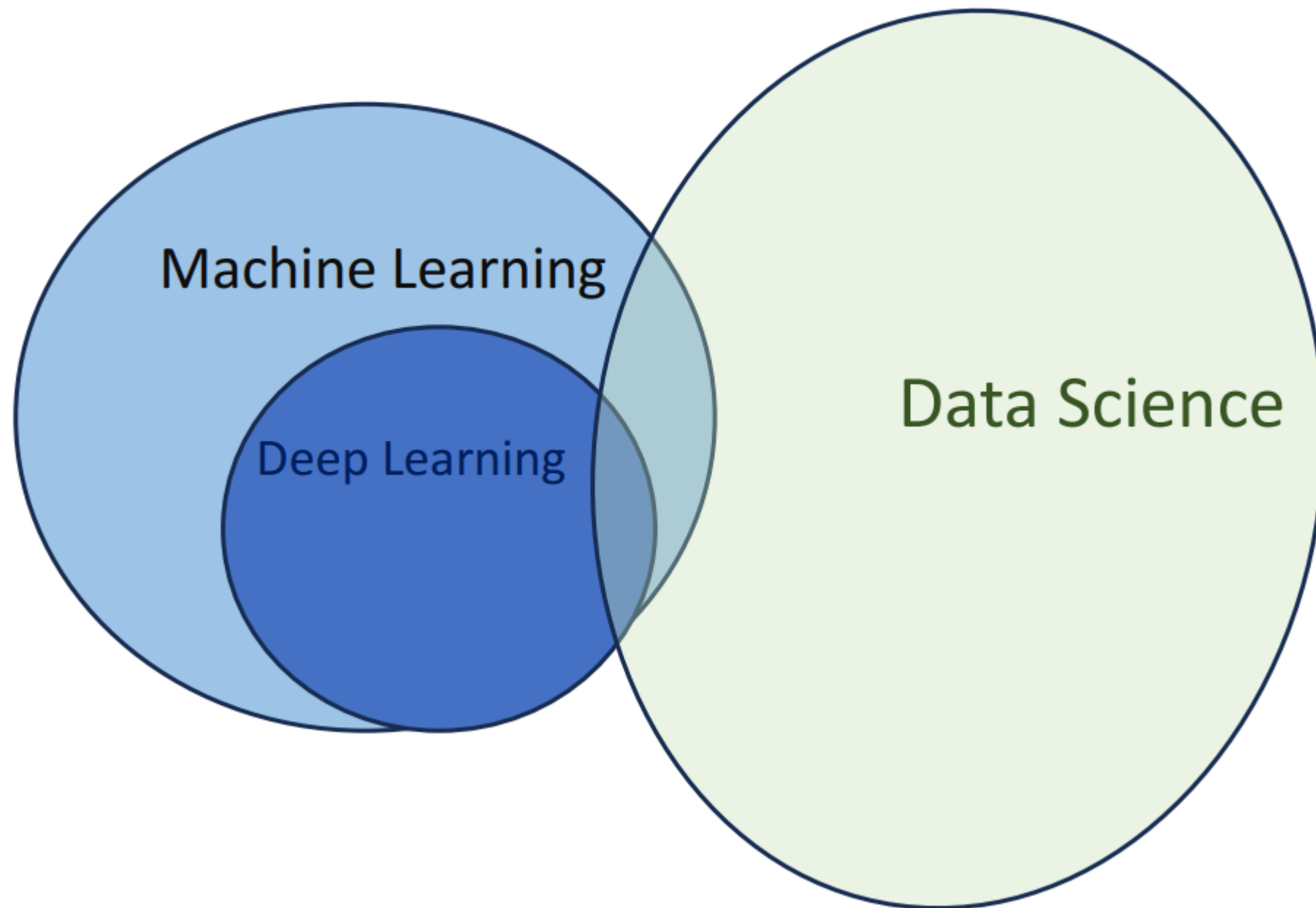
Machine learning workflows, similar to data science

- Differences

- Post Analysis

Data scientists typically interpret and report their findings, whereas machine learning engineers focus on the model's deployment, monitoring, and maintaining in a production environment.

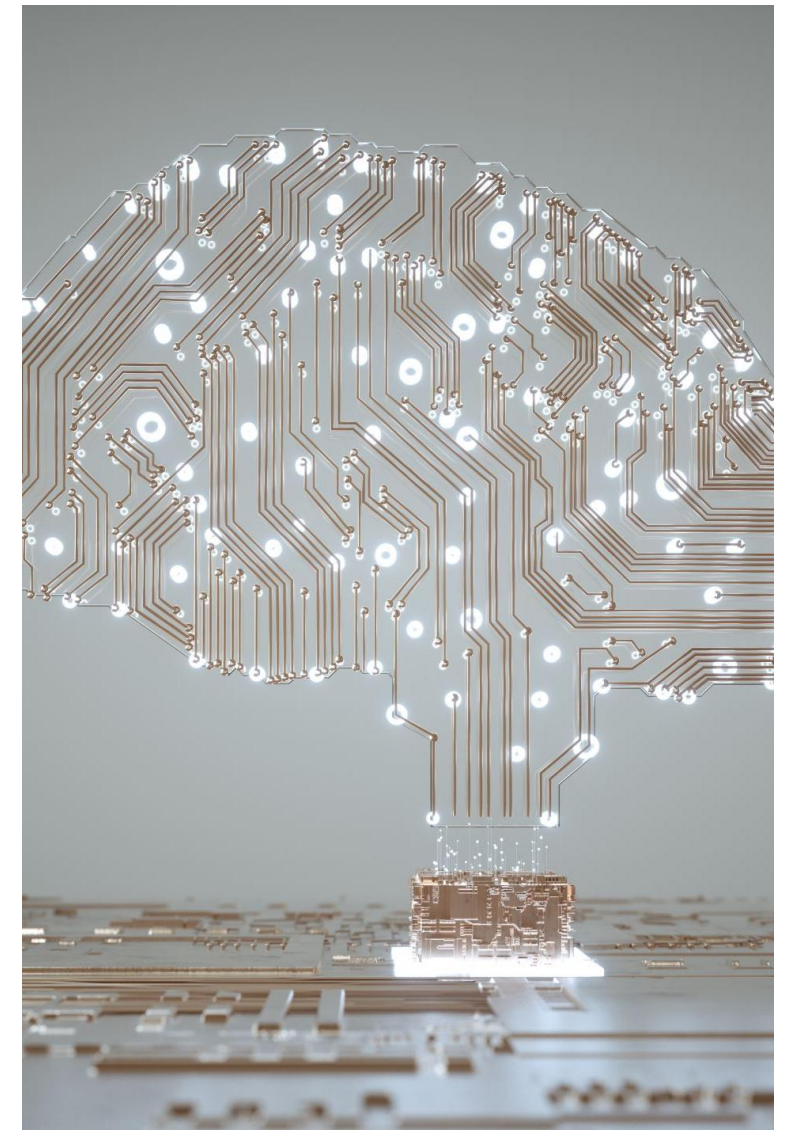
Data Science and Machine Learning



Artificial Intelligence and Data Science

Artificial Intelligence

- **Artificial intelligence** makes machines seem like they have human intelligence.
- The goal of AI is to:
 - Mimic the human and
 - Create systems that can function intelligently and independently.

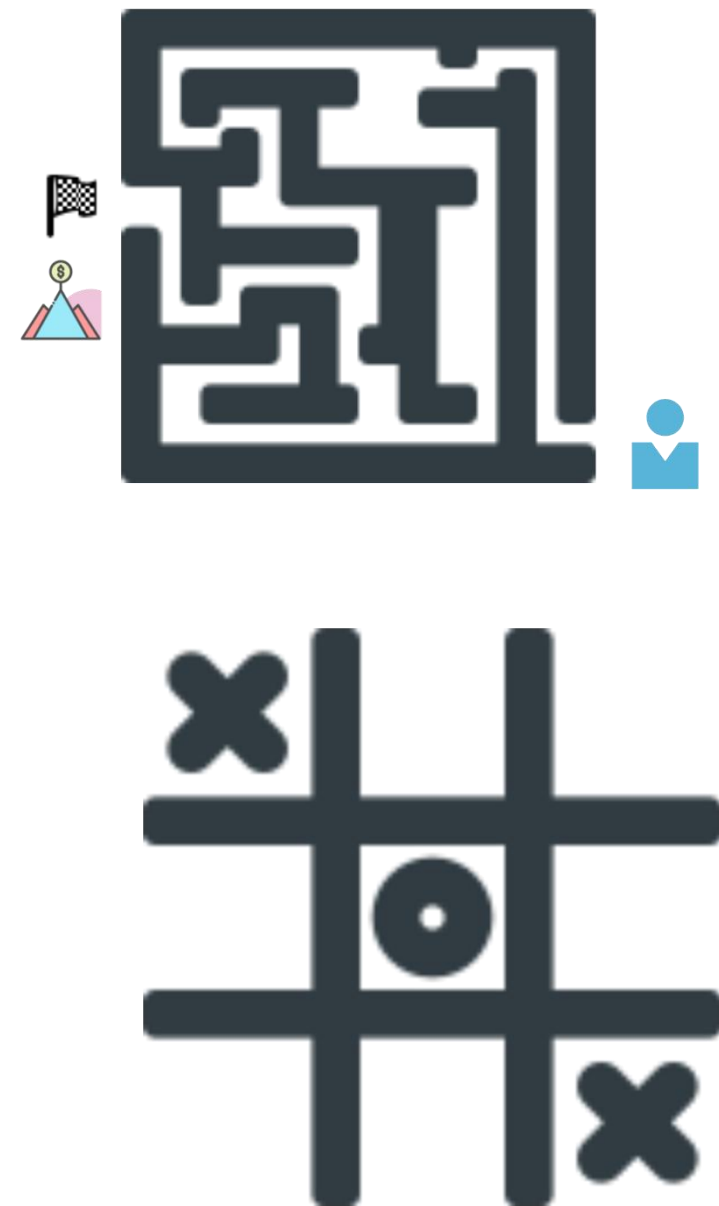


AI and Data Science

- Tasks that constitute AI
 - Problem Solving
 - Knowledge Representation and Reasoning
 - Decision Making
 - Communication, perception and Actuation

Problem Solving

- Problem Solving
 - Maze game
 - Tic tac toe
- (No data no modelling)
- (Need efficient algo. A*, DFS/BFS)



Knowledge Representation & Reasoning

- Knowledge Representation & Reasoning
 - Eight Queen's problem
 - complex game rules

(Write rules and apply reasoning)

(No data no modelling)

(Need propositional and first order logic.)

Decision Making

- Decision Making
 - ✓ Expert Systems (when rules are well known)
(Rules given by experts)
(Rules encoded using knowledge representation)
(Execution of rules by a program)
 - ✓ Limitation of Expert Systems
 - Rules may be too complex
 - Rules may be inexpressible
 - Rules may be unknown (specially for new diseases)
 - ✓ Here we can take help from Machine Learning, Reinforcement Learning, Deep Learning (No need to write rules).

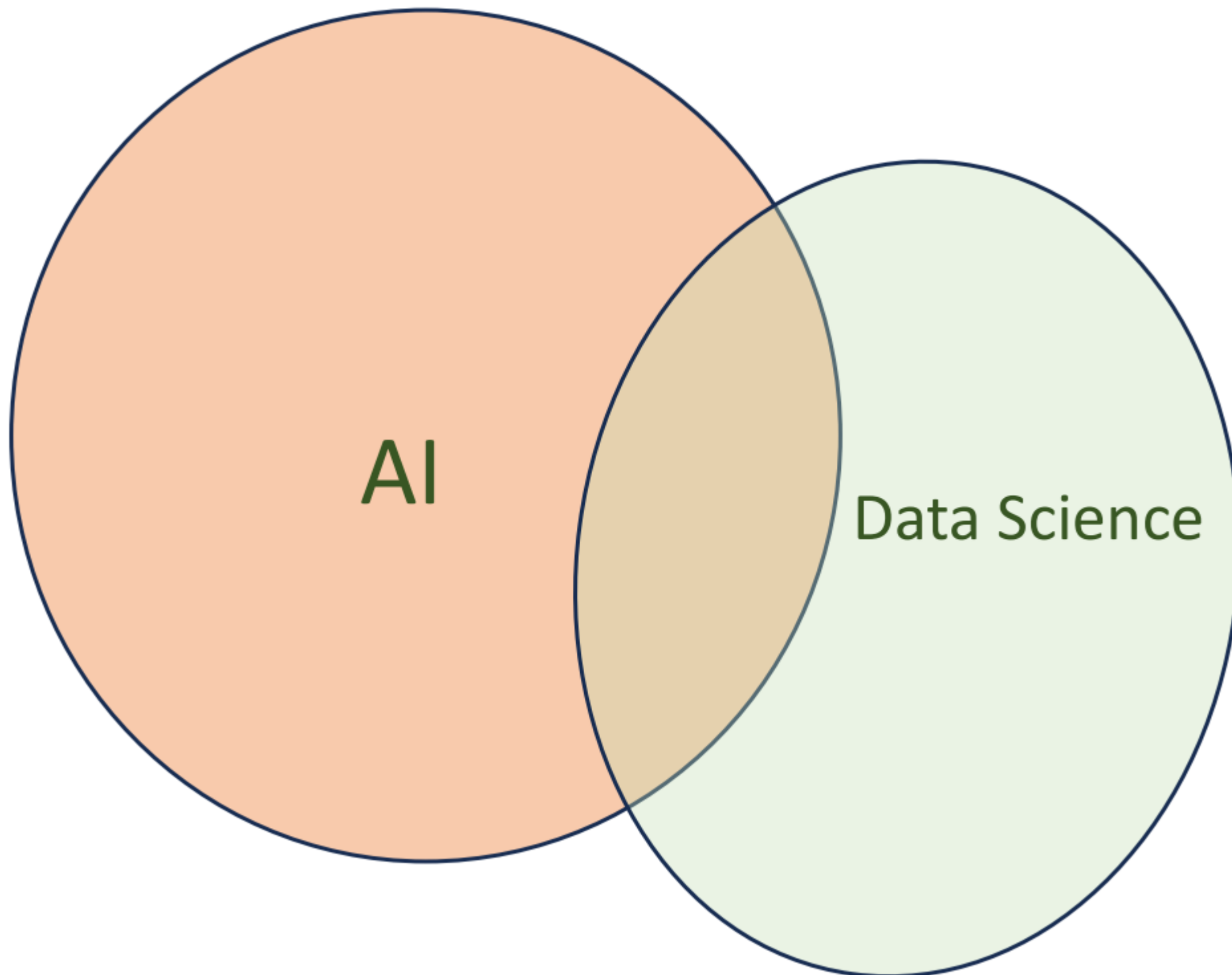
Communication, perception and Actuation

- Communication
 - Traditionally AI agents (or robots) are required to communicate with humans.
 - The agent should have capability of Natural Language Processing
 - Modern NLP is data driven and ML/DL is best choice for NLP
- (**Example:** An agent can perform voice analysis using Machine learning/Deep Learning)

Communication, perception and Actuation

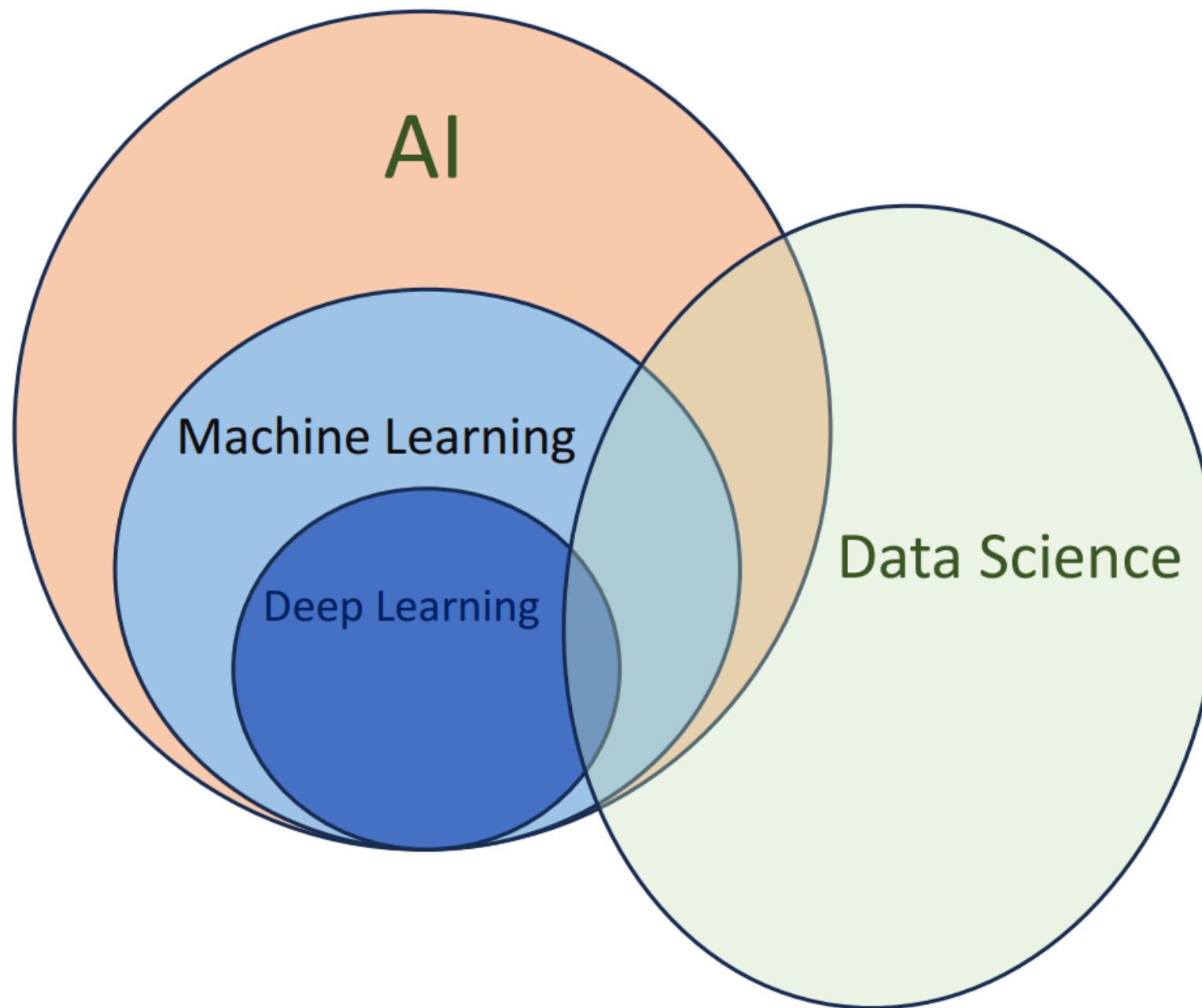
- Perception
 - How agent perceive the environment (i.e., how agent observe around its surrounding)
 - It gives ability of vision and speech
 - Modern CV (computer vision) and speech are data driven
- Actuation
 - Performing some action by an agent (i.e robot)
 - Reinforcement Learning
 - This is data driven now.

AI vs Data Science



	Data Science	AI	Machine Learning
Focus	Extracting insight from structured and unstructured data for decision-making and planning.	Enable computers to accomplish complex intellectual tasks like humans.	Provide a way to synthesize data, learn from it and use the insights to improve over time.
Application	Solving Business problems using descriptive and predictive analytic. Examples: Disease prediction, Financial analysis	Perform tasks like humans by learning, reasoning, and self-correction. Examples: Robots, Voice assistants, Online gaming.	Extract knowledge from data to learn from that data and make predictions. Examples: recommendation system, Health monitoring.

AI vs Machine Learning vs Data Science



Summary

- In what way Data Science is related with Machine Learning.
- The way AI and Data Science are related
- Finally, the relation between AI, Machine Learning and Data Science.